



PUCP

TEL141 - INGENIERÍA DE REDES CLOUD

Reporte Final

Laboratorio N°:

Semestre:

Tema:

AUTOMATIZACIÓN DE
ORQUESTACIÓN/APROVISIONAMIENTO DE VMs EN UN
CLÚSTER

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REPORTE FINAL

ACTIVIDAD N° 1: red VLAN sin DHCP y con salida a internet (1 pt)

- Adjunte evidencia de lo ejecutado y las salidas en el servidor con rol cliente.

Responder aquí.

```
[OK] create_vm.sh finalizado. VM vm1_vlan200 disponible por VNC en el puerto 5902
[i] VM: vm2_vlan100 | VLAN: 100 | VNC: 0.0.0.0:1 | MAC: 22:12:03:00:64:02
[=] Imagen base ya existe: /home/ubuntu/images/cirros-0.5.1-x86_64-disk.img
[+] Limpiando restos previos de la VM...
[+] Deteniendo QEMU previo (PID 160708)
[+] Creando overlay /home/ubuntu/images/vm2_vlan100.qcow2 sobre imagen base
[+] Creando interfaz TAP vm2_vlan100_tap
[+] Lanzando QEMU en background (VNC :1)
qemu-system-x86_64: warning: host doesn't support requested feature: CPUID.80000001H:ECX.svm [bit 2]
[=] QEMU corriendo con PID 161085
[+] Conectando vm2_vlan100_tap a br-int con tag 100
[OK] create_vm.sh finalizado. VM vm2_vlan100 disponible por VNC en el puerto 5901
[i] VM: vm2_vlan200 | VLAN: 200 | VNC: 0.0.0.0:2 | MAC: 22:12:03:00:c8:02
[=] Imagen base ya existe: /home/ubuntu/images/cirros-0.5.1-x86_64-disk.img
[+] Limpiando restos previos de la VM...
[+] Deteniendo QEMU previo (PID 160843)
[+] Creando overlay /home/ubuntu/images/vm2_vlan200.qcow2 sobre imagen base
[+] Creando interfaz TAP vm2_vlan200_tap
[+] Lanzando QEMU en background (VNC :2)
qemu-system-x86_64: warning: host doesn't support requested feature: CPUID.80000001H:ECX.svm [bit 2]
[=] QEMU corriendo con PID 161217
[+] Conectando vm2_vlan200_tap a br-int con tag 200
[OK] create_vm.sh finalizado. VM vm2_vlan200 disponible por VNC en el puerto 5902
[6/6] Habilitando salida a Internet para VLAN 100 y 200
[i] Eliminando enrutamiento entre VLAN 100 (gw_vlan100) y VLAN 200 (gw_vlan200)
[OK] no_routing_networks.sh finalizado (VLAN 100 <-x-> VLAN 200)
[i] Habilitando salida a Internet para VLAN 100 (red 192.168.0.0/24) vía ens3
[OK] internet_to_network.sh finalizado para VLAN 100
[i] Habilitando salida a Internet para VLAN 200 (red 192.168.2.0/24) vía ens3
[OK] internet_to_network.sh finalizado para VLAN 200
[READY] Actividad 2 desplegada
```

SERVIDOR	VM	VLAN	IP	GW	VNC
SERVER1	vm1_vlan100	100	192.168.0.10	192.168.0.1	:1 (5901)
SERVER2	vm2_vlan100	100	192.168.0.11	192.168.0.1	:2 (5902)
SERVER1	vm1_vlan200	200	192.168.2.10	192.168.2.1	:1 (5901)
SERVER2	vm2_vlan200	200	192.168.2.11	192.168.2.1	:2 (5902)

CONFIGURAR: VMs necesitan IPs y GW vía VNC.
ubuntu@server4:~/TEL141_L4_20221203/actividades\$

- Muestre la IP asignada al contenedor en la VLAN 100 en el servidor 1 y a la VM en la VLAN 100 en el servidor 2.

Server1:

```
$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 22:12:03:00:64:01 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.10/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::2012:3ff:fe00:6401/64 scope link
        valid_lft forever preferred_lft forever
$
```

Server2:

```
$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 22:12:03:00:64:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.11/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::2012:3ff:fe00:6402/64 scope link
        valid_lft forever preferred_lft forever
```

- Conectividad capa 2 (arping) y capa 3 (ping) con el default gateway

Server 1:

```
$ sudo arping 192.168.0.1
ARPING 192.168.0.1 from 192.168.0.10 eth0
Unicast reply from 192.168.0.1 [9a:5e:6b:76:72:8d] 7.219ms
Unicast reply from 192.168.0.1 [9a:5e:6b:76:72:8d] 3.773ms
Unicast reply from 192.168.0.1 [9a:5e:6b:76:72:8d] 1.290ms
Unicast reply from 192.168.0.1 [9a:5e:6b:76:72:8d] 1.190ms
^CSent 4 probe(s) (1 broadcast(s))
Received 4 response(s) (0 request(s), 0 broadcast(s))
$ ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
64 bytes from 192.168.0.1: seq=0 ttl=64 time=8.066 ms
64 bytes from 192.168.0.1: seq=1 ttl=64 time=2.422 ms
64 bytes from 192.168.0.1: seq=2 ttl=64 time=2.509 ms
64 bytes from 192.168.0.1: seq=3 ttl=64 time=2.348 ms
^C
--- 192.168.0.1 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 2.348/3.836/8.066 ms
$
```

Server 2:

```
$ sudo arping 192.168.0.1
ARPING 192.168.0.1 from 192.168.0.10 eth0
Unicast reply from 192.168.0.1 [9a:5e:6b:76:72:8d] 5.244ms
Unicast reply from 192.168.0.1 [9a:5e:6b:76:72:8d] 3.021ms
Unicast reply from 192.168.0.1 [9a:5e:6b:76:72:8d] 0.042ms
Unicast reply from 192.168.0.1 [9a:5e:6b:76:72:8d] 0.043ms
^CSent 4 probe(s) (1 broadcast(s))
Received 4 response(s) (0 request(s), 0 broadcast(s))
$ ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1): 56 data bytes
64 bytes from 192.168.0.1: seq=0 ttl=64 time=5.836 ms
64 bytes from 192.168.0.1: seq=1 ttl=64 time=2.091 ms
64 bytes from 192.168.0.1: seq=2 ttl=64 time=2.055 ms
64 bytes from 192.168.0.1: seq=3 ttl=64 time=2.281 ms
^C
--- 192.168.0.1 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 2.055/3.065/5.836 ms
$
```

- Validar conectividad con internet desde el contenedor en la VLAN 100 en el servidor 1.

```
# Responder aquí.  
$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8): 56 data bytes  
64 bytes from 8.8.8.8: seq=0 ttl=110 time=43.726 ms  
64 bytes from 8.8.8.8: seq=1 ttl=110 time=37.679 ms  
64 bytes from 8.8.8.8: seq=2 ttl=110 time=37.195 ms  
64 bytes from 8.8.8.8: seq=3 ttl=110 time=37.298 ms  
^C  
--- 8.8.8.8 ping statistics ---  
4 packets transmitted, 4 packets received, 0% packet loss  
round-trip min/avg/max = 37.195/38.974/43.726 ms  
$
```

- Validar conectividad con internet desde la VM en la VLAN 100 en el servidor 2.

```
# Responder aquí.  
$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8): 56 data bytes  
64 bytes from 8.8.8.8: seq=0 ttl=110 time=37.237 ms  
64 bytes from 8.8.8.8: seq=1 ttl=110 time=37.663 ms  
64 bytes from 8.8.8.8: seq=2 ttl=110 time=37.280 ms  
64 bytes from 8.8.8.8: seq=3 ttl=110 time=37.354 ms  
^C  
--- 8.8.8.8 ping statistics ---  
4 packets transmitted, 4 packets received, 0% packet loss  
round-trip min/avg/max = 37.237/37.383/37.663 ms  
$
```

ACTIVIDAD N° 2: red con DHCP y sin salida a internet (3 pt)

- Adjunte evidencia de lo ejecutado y las salidas en el servidor con rol cliente.

```
[OK] create_vm.sh finalizado. VM vm1_vlan200 disponible por VNC en el puerto 5902
[i] VM: vm2_vlan100 | VLAN: 100 | VNC: 0.0.0.0:1 | MAC: 22:12:03:00:64:02
[=] Imagen base ya existe: /home/ubuntu/images/cirros-0.5.1-x86_64-disk.img
[+] Limpiando restos previos de la VM...
[+] Deteniendo QEMU previo (PID 161085)
[+] Creando overlay /home/ubuntu/images/vm2_vlan100.qcow2 sobre imagen base
[+] Creando interfaz TAP vm2_vlan100_tap
[+] Lanzando QEMU en background (VNC :1)
qemu-system-x86_64: warning: host doesn't support requested feature: CPUID.80000001H:ECX.svm [bit 2]
[=] QEMU corriendo con PID 161678
[+] Conectando vm2_vlan100_tap a br-int con tag 100
[OK] create_vm.sh finalizado. VM vm2_vlan100 disponible por VNC en el puerto 5901
[i] VM: vm2_vlan200 | VLAN: 200 | VNC: 0.0.0.0:2 | MAC: 22:12:03:00:c8:02
[=] Imagen base ya existe: /home/ubuntu/images/cirros-0.5.1-x86_64-disk.img
[+] Limpiando restos previos de la VM...
[+] Deteniendo QEMU previo (PID 161217)
[+] Creando overlay /home/ubuntu/images/vm2_vlan200.qcow2 sobre imagen base
[+] Creando interfaz TAP vm2_vlan200_tap
[+] Lanzando QEMU en background (VNC :2)
qemu-system-x86_64: warning: host doesn't support requested feature: CPUID.80000001H:ECX.svm [bit 2]
[=] QEMU corriendo con PID 161813
[+] Conectando vm2_vlan200_tap a br-int con tag 200
[OK] create_vm.sh finalizado. VM vm2_vlan200 disponible por VNC en el puerto 5902
[6/6] Asegurando aislamiento de VLANs y bloqueo de internet
[i] Eliminando enrutamiento entre VLAN 100 (gw_vlan100) y VLAN 200 (gw_vlan200)
[OK] no_routing_networks.sh finalizado (VLAN 100 <-x-> VLAN 200)
[i] Eliminando salida a Internet para VLAN 100 (red 192.168.0.0/24) vía ens3
[OK] no_internet_to_network.sh finalizado para VLAN 100
[i] Eliminando salida a Internet para VLAN 200 (red 192.168.2.0/24) vía ens3
[OK] no_internet_to_network.sh finalizado para VLAN 200
[READY] Actividad 3 desplegada
```

SERVIDOR	VM	VLAN	IP	GW	VNC
SERVER1	vm1_vlan100	100	192.168.0.X/24	192.168.0.1	:1 (5901)
SERVER2	vm2_vlan100	100	192.168.0.X/24	192.168.0.1	:2 (5902)
SERVER1	vm1_vlan200	200	192.168.2.X/24	192.168.2.1	:1 (5901)
SERVER2	vm2_vlan200	200	192.168.2.X/24	192.168.2.1	:2 (5902)

VERIFICAR: VMs obtienen IPs y GW por DHCP, ver via VNC.
ubuntu@server4:~/TEL141_L4_20221203/actividades\$

- Muestre la IP obtenido por el contenedor de la VLAN 200 en el servidor 1.
 - Solicitar una IP mediante DHCP

```
# Responder aquí.
$ sudo dhcpcd -i eth0
dhcpcd: started, v1.29.3
WARN: '/usr/share/dhcpcd/default.script' should not be used in cirros. Replaced
by cirros-dhcpcd.
dhcpcd: sending discover
dhcpcd: sending select for 192.168.2.68
dhcpcd: lease of 192.168.2.68 obtained, lease time 43200
WARN: '/usr/share/dhcpcd/default.script' should not be used in cirros. Replaced
by cirros-dhcpcd.
$
```

- Mostrar la IP obtenida. Asimismo, muestre mac de la interfaz.

```
# Responder aquí.
```

```
$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 22:12:03:00:c8:01 brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.68/24 brd 192.168.2.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::2012:3fff:fe00:c801/64 scope link
        valid_lft forever preferred_lft forever
$
```

- Muestre el default gateway obtenido

```
# Responder aquí.
$ ip ro
default via 192.168.2.1 dev eth0
192.168.2.0/24 dev eth0 scope link src 192.168.2.68
$
```

- Conectividad capa 2 (arping) y capa 3 (ping) con el default gateway

```
# Responder aquí.
$ sudo arping 192.168.2.1
ARPING 192.168.2.1 from 192.168.2.68 eth0
Unicast reply from 192.168.2.1 [32:0c:2e:6b:66:13] 4.602ms
Unicast reply from 192.168.2.1 [32:0c:2e:6b:66:13] 2.453ms
Unicast reply from 192.168.2.1 [32:0c:2e:6b:66:13] 0.993ms
Unicast reply from 192.168.2.1 [32:0c:2e:6b:66:13] 1.126ms
^CSent 4 probe(s) (1 broadcast(s))
Received 4 response(s) (0 request(s), 0 broadcast(s))
$ ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1): 56 data bytes
64 bytes from 192.168.2.1: seq=0 ttl=64 time=6.750 ms
64 bytes from 192.168.2.1: seq=1 ttl=64 time=2.408 ms
64 bytes from 192.168.2.1: seq=2 ttl=64 time=2.335 ms
64 bytes from 192.168.2.1: seq=3 ttl=64 time=1.955 ms
^C
--- 192.168.2.1 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.955/3.362/6.750 ms
$
```

- En la máquina virtual en VLAN 200 instanciada en el servidor 2, realice lo siguiente y adjunte la salida al ejecutar sus comando de lo solicitado:

- Solicitar una IP mediante DHCP

```
# Responder aquí.
```

```
$ sudo udhcpd -i eth0
udhcpd: started, v1.29.3
WARN: '/usr/share/udhcpd/default.script' should not be used in cirros. Replaced
by cirros-dhcpd.
udhcpd: sending discover
udhcpd: sending select for 192.168.2.69
udhcpd: lease of 192.168.2.69 obtained, lease time 43200
WARN: '/usr/share/udhcpd/default.script' should not be used in cirros. Replaced
by cirros-dhcpd.
$
```

- Mostrar la IP obtenida. Asimismo, muestre mac de la interfaz.

```
# Responder aquí.
$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 22:12:03:00:c8:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.69/24 brd 192.168.2.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::2012:3ff:fe00:c802/64 scope link
        valid_lft forever preferred_lft forever
$
```

- Muestre el default gateway obtenido

```
# Responder aquí.
$ sudo ip ro
default via 192.168.2.1 dev eth0
192.168.2.0/24 dev eth0 scope link src 192.168.2.69
$
```

- Conectividad capa 2 (arping) y capa 3 (ping) con el default gateway

```
# Responder aquí.
$ sudo arping 192.168.2.1
ARPING 192.168.2.1 from 192.168.2.69 eth0
Unicast reply from 192.168.2.1 [32:0c:2e:6b:66:13] 8.524ms
Unicast reply from 192.168.2.1 [32:0c:2e:6b:66:13] 4.019ms
Unicast reply from 192.168.2.1 [32:0c:2e:6b:66:13] 0.042ms
Unicast reply from 192.168.2.1 [32:0c:2e:6b:66:13] 0.046ms
^C Sent 4 probe(s) (1 broadcast(s))
Received 4 response(s) (0 request(s), 0 broadcast(s))
$ ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1): 56 data bytes
64 bytes from 192.168.2.1: seq=0 ttl=64 time=7.146 ms
64 bytes from 192.168.2.1: seq=1 ttl=64 time=2.507 ms
64 bytes from 192.168.2.1: seq=2 ttl=64 time=2.070 ms
64 bytes from 192.168.2.1: seq=3 ttl=64 time=2.357 ms
^C
--- 192.168.2.1 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 2.070/3.520/7.146 ms
$
```

ACTIVIDAD N° 3: Ruteo entre redes vlan (2 pt)

- Adjunte evidencia de lo ejecutado y las salidas en el servidor con rol cliente.

```
# Responder aquí.

[+] Creando overlay /home/ubuntu/images/vm2_vlan100.qcow2 sobre imagen base
[+] Creando interfaz TAP vm2_vlan100_tap
[+] Lanzando QEMU en background (VNC :1)
qemu-system-x86_64: warning: host doesn't support requested feature: CPUID.80000001H:ECX.svm [bit 2]
[+] QEMU corriendo con PID 162496
[+] Conectando vm2_vlan100_tap a br-int con tag 100
[OK] create_vm.sh finalizado. VM vm2_vlan100 disponible por VNC en el puerto 5901
[i] VM: vm2_vlan200 | VLAN: 200 | VNC: 0.0.0.0:2 | MAC: 22:12:03:00:c8:02
[+] Imagen base ya existe: /home/ubuntu/images/cirros-0.5.1-x86_64-disk.img
[+] Limpiando restos previos de la VM...
[+] Deteniendo QEMU previo (PID 162198)
[+] Creando overlay /home/ubuntu/images/vm2_vlan200.qcow2 sobre imagen base
[+] Creando interfaz TAP vm2_vlan200_tap
[+] Lanzando QEMU en background (VNC :2)
qemu-system-x86_64: warning: host doesn't support requested feature: CPUID.80000001H:ECX.svm [bit 2]
[+] QEMU corriendo con PID 162630
[+] Conectando vm2_vlan200_tap a br-int con tag 200
[OK] create_vm.sh finalizado. VM vm2_vlan200 disponible por VNC en el puerto 5902
[6/7] Asegurando que NO haya NAT (Act4 no lleva Internet)
[i] Eliminando salida a Internet para VLAN 100 (red 192.168.0.0/24) vía ens3
[OK] no_internet_to_network.sh finalizado para VLAN 100
[i] Eliminando salida a Internet para VLAN 200 (red 192.168.2.0/24) vía ens3
[OK] no_internet_to_network.sh finalizado para VLAN 200
[7/7] Habilitando enrutamiento entre VLAN 100 y VLAN 200
[i] Habilitando enrutamiento entre VLAN 100 (gw_vlan100) y VLAN 200 (gw_vlan200)
[OK] routing_networks.sh finalizado (VLAN 100 <-> VLAN 200)
[i] Eliminando salida a Internet para VLAN 100 (red 192.168.0.0/24) vía ens3
[OK] no_internet_to_network.sh finalizado para VLAN 100
[i] Eliminando salida a Internet para VLAN 200 (red 192.168.2.0/24) vía ens3
[OK] no_internet_to_network.sh finalizado para VLAN 200
[READY] Actividad 4 desplegada
```

SERVIDOR	VM	VLAN	IP	GW	VNC
SERVER1	vm1_vlan100	100	192.168.0.X/24	192.168.0.1	:1 (5901)
SERVER2	vm2_vlan100	100	192.168.0.X/24	192.168.0.1	:2 (5902)
SERVER1	vm1_vlan200	200	192.168.2.X/24	192.168.2.1	:1 (5901)
SERVER2	vm2_vlan200	200	192.168.2.X/24	192.168.2.1	:2 (5902)

```
VERIFICAR: VMs obtienen IPs y GW por DHCP, ver via VNC.
ubuntu@server4:~/TEL141_14_20221203/actividades$
```

- Valide la conectividad entre los contenedores y máquinas virtuales en la VLAN 100 y 200:

VM VLAN 100:

```

$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 22:12:03:00:64:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.86/24 brd 192.168.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::2012:3ff:fe00:6402/64 scope link
        valid_lft forever preferred_lft forever
$ ping 192.168.2.69
PING 192.168.2.69 (192.168.2.69): 56 data bytes
64 bytes from 192.168.2.69: seq=0 ttl=63 time=14.124 ms
64 bytes from 192.168.2.69: seq=1 ttl=63 time=3.936 ms
64 bytes from 192.168.2.69: seq=2 ttl=63 time=3.888 ms
64 bytes from 192.168.2.69: seq=3 ttl=63 time=4.051 ms
^C
--- 192.168.2.69 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 3.888/6.499/14.124 ms
$

```

VM VLAN 200:

```

$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 22:12:03:00:c8:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.69/24 brd 192.168.2.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::2012:3ff:fe00:c802/64 scope link
        valid_lft forever preferred_lft forever
$ ping 192.168.0.86
PING 192.168.0.86 (192.168.0.86): 56 data bytes
64 bytes from 192.168.0.86: seq=0 ttl=63 time=21.652 ms
64 bytes from 192.168.0.86: seq=1 ttl=63 time=4.634 ms
64 bytes from 192.168.0.86: seq=2 ttl=63 time=3.926 ms
64 bytes from 192.168.0.86: seq=3 ttl=63 time=3.997 ms
^C
--- 192.168.0.86 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 3.926/8.552/21.652 ms
$ _

```